Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	***Distributed Model Predictive Path Integral Control for Swarms and Cooperative Robotics.***				
Korea's 12 National Strategic Technology Fields	**Semiconductor/Display (), Secondary Battery (), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (v), Quantum ( )**				
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**			**Technology Field 2 (**Large Category**)**	
	Advanced Robotics				
	Medium Category		Small Category	Medium Category	Small Category
	Advanced Mobility		***Aerospace/Marine***		
Concept of Proposed Technology	*The proposed technology aims to control multi-agent robotic and autonomous systems performing cooperative tasks in the presence of uncertainty. These tasks will include surveillance and monitoring, transportation of heavy objects, collaborative construction and others. The final product will be a software architecture that supports distributed decision-making for multi-agentic autonomous systems. From an algorithmic perspective the technology will rely on Distributed Differential Dynamic Programming, as well as Model Predictive Path Integral (MPPI) control and its extensions to distributed optimization. MPPI and the Distributed Differential Dynamic Programming architectures are flagship algorithms of PI's group the Autonomous Control and Decision Systems laboratory at Georgia Institute of Technology.*				

	MPPI is a very mature stochastic optimization framework with a broad range of applications and use cases in robotics and autonomy. In addition to its maturity, the entry point for using, extending and deploying this MPPI is very low. This is attributed to the flexibility that the MPPI framework has in handling different types of robotics systems with different levels of complexity and task requirements. MPPI is fully parallelizable and can run in real time when using a GPU. It can handle highly nonlinear and non-smooth dynamics and cost functions. *The Nested and Merged Distributed Differential Dynamic Programming architectures are state of the art distributed trajectory optimization methods for large-scale swarm control. Both algorithms have substantial scaling capabilities and can handle different types of dynamics and cost functions assuming differentiability and smoothness. The implementation of Distributed Differential Dynamic Programming algorithms does not require a GPU.* *Given the maturity of the work on MPPI and its successes in field robotics for single-agent problems, the next step for this planning technology is to scale MPPI to multi-agent systems. This is not a very high-risk project since all the moving pieces are there in terms of algorithms and technologies to design an architecture that can support multi-agent operations at scale. The distributed version of MPPI will be benchmarked against Distributed Differential Dynamic Programming algorithms and the best distributed architecture will be chosen for real world robotic experiments and testing.* *The final product should be able to handle different types of robots including Unmanned Aerial, Sea and Terrestrial Vehicles as well as different types of cooperative tasks.*
Rationale for Joint R&D Project	*Multi-agent robotics systems and swarms are becoming more and more prevalent in different industries from constructions and transportation to defense related domains and tasks. Initiating a project on distributed MPPI for multi-agent robotics is very timely due to the maturity of planning technologies such as MPPI, the advancement in low energy GPUs for online computation, and the ability to design and build low cost UAVs at scale. From an industrial, economic and geo-political perspective there is the need for the deployment of large numbers of UAV for performing dangerous, highly sensitive and highly safety-critical tasks. Therefore the need and the market is there for products that can support distributed capabilities for teams of robots at scale.*
Key Words	1. *Provide key terms that describe the proposed technology and its main features (min 3, max 5)* 2. *Distributed and scalable decision-making under uncertainty* 3. *Robust operation in GPS denied environments* 4. *Fast adaptation to changing environments* 5. *Intelligent cooperation.*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]		**Start of the Project**	Level ( )	**End of the Project**	Level ( )
R&D Objectives		*The development of distributed algorithms for autonomous operation of large-scale teams of robots and fleets of autonomous vehicles. The technology should be applicable to different types of robots and for the different collaborative tasks including formation, heavy object transportation and delivery, cooperative construction, cooperative defense.*			
R&D Target Specifications		*The criteria to measure success in the proposed research include:* - *<u>Scalability of the Optimization:</u> Number of vehicles that can be controlled in parallel in a GPS denied environment.* - *<u>Adaptability and Response Speed:</u> Success rates in transporting objects in different environments for different levels of non-stationarity and disturbances.* - *<u>Robustness to Partial Observability:</u> Performance degradation or improvement as a function on on-board sensing capabilities.* - *<u>Redundancy capabilities:</u> Success rates in performing cooperative tasks when some agents from the team fail or become uncontrollable.*			
R&D Plan		*From a milestone perspective below is a tentative list of milestones:* - ***Milestone 1 - Year 1*** - *Distributed MPPI and Distributed DDP under full observability using a VICON system..* - *Benchmarking on simulations and deployment on drones or terrestrial vehicles using indoor facilities at GT. The goal is to choose the best architecture for the next milestone* - ***Tasks**: Vehicle formation, object transportation or other cooperative tasks.* ***Milestone 2 - Year 2:*** - *Distributed MPPI or DDP under partial observability with Kalman Filter in the Loop using GPS. Deployment on drones or terrestrial vehicles.* - *Outdoor hardware evaluations* - ***Tasks:** Vehicle formation, object transportation or other cooperative tasks.* ***Milestone 3 - Year 3:*** - *Distributed MPPI or DDP under partial observability with Kalman Filter in*			

		a GPS denied environment.Deployment on drones or terrestrial vehicles. - *Outdoor hardware evaluations* - ***Tasks:*** *Vehicle formation, object transportation or other cooperative tasks.* These milestones above apply also to use-cases of autonomy that include sea vehicles. The deployment of such systems is definitely challenging from a logistic perspective. However, given the tradition that South Korea has in terms of developing naval technologies, the hope is that there will be startups and stake-holders in South Korea that would be interested in developing sea swarm capabilities.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*The ACDS lab at Georgia Tech will be responsible for the development of the software for the distributed planning and state estimation modules. This includes the following items:* - *Development of distributed Model Predictive Path Integral Control planning architectures.* - *Comparisons with alternative approaches such as Distributed Differential Dynamic Programming.* - *Development of distributed State Estimation pipelines* - *Integration and testing in software, simulation and indoor testing facilities.* *The ACDS lab already has two first year PhD students from South Korea who work on the domain of distributed planning and state estimation for teams of robotics systems. If successful the funds from this research will support the two Korean students.*
Expected R&R of Korean SME Partner	*The Korean SME partner will be responsible for:* - - *Design of the hardware with specific sensing capabilities or provide the hardware if this already exists.* - *Integration of the distributed planning and state estimation software for testing at scale.* - *Testing and validation of the entire pipeline in indoor and outdoor facilities for different tasks and different scales of the swarm.* - *Rigorous benchmarking for meeting success criteria.*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*The expected outcomes will include:* 1. *Distributed trajectory optimization software package with decision-making architectures that can be commercializable.* 2. *Distributed state estimation software package with state-estimation architectures that can be commercializable* 3. *End-to-end distributed autonomy stack architecture that includes distributed sensing, planning and control modules. The entire pipeline can be a commercial product.* 4. *The software packages will be applicable to different multi-agent robotic systems and tasks and it should be hardware and/or robot agnostic.* 5. *The impact of the proposed work is substantial given the need to develop technologies that can handle the complexity of scale of multi-agentic systems. Therefore while the starting point of this work is on robotic systems, it can be extended to multi-agentic systems in other areas of science and industry.*

6. References

(1) G. Williams, P. Drews, B. Goldfain, J. M. Rehg and E. A. Theodorou, "Aggressive driving with model predictive path integral control," *2016 IEEE International Conference on Robotics and Automation (ICRA)*, Stockholm, Sweden, 2016, pp. 1433-1440, doi: 10.1109/ICRA.2016.7487277.

(2) G. Williams, P. Drews, B. Goldfain, J. M. Rehg and E. A. Theodorou, "Information-Theoretic Model Predictive Control: Theory and Applications to Autonomous Driving," in *IEEE Transactions on Robotics*, vol. 34, no. 6, pp. 1603-1622, Dec. 2018, doi: 10.1109/TRO.2018.2865891

(3) M. S. Gandhi, B. Vlahov, J. Gibson, G. Williams and E. A. Theodorou, "Robust Model Predictive Path Integral Control: Analysis and Performance Guarantees," in *IEEE Robotics and Automation Letters*, vol. 6, no. 2, pp. 1423-1430, April 2021, doi: 10.1109/LRA.2021.3057563.

(4) B. Vlahov, J. Gibson, D. D. Fan, P. Spieler, A. -a. Agha-mohammadi and E. A. Theodorou, "Low Frequency Sampling in Model Predictive Path Integral Control," in *IEEE Robotics and Automation Letters*, vol. 9, no. 5, pp. 4543-4550, May 2024

(5) A. D. Saravanos, Y. Aoyama, H. Zhu and E. A. Theodorou, "Distributed Differential Dynamic Programming Architectures for Large-Scale Multiagent Control," in *IEEE Transactions on Robotics*, vol. 39, no. 6, pp. 4387-4407, Dec. 2023,

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization